# CoStream: Compositional Multi-Stream Framework for Multi-Modal Manipulation

Haonan Chen[1], Yuxiang Ma[2], Stephen Tian[3], Xiaoshen Han[1], Wenlong Huang[3],
Yunzhu Li[3], Jiajun Wu[3], Edward H Adelson[2], Yilun Du[1]
[1] Harward University [2] Massachusetts Institute of Technology [3] Stanford University [4] Columbia University

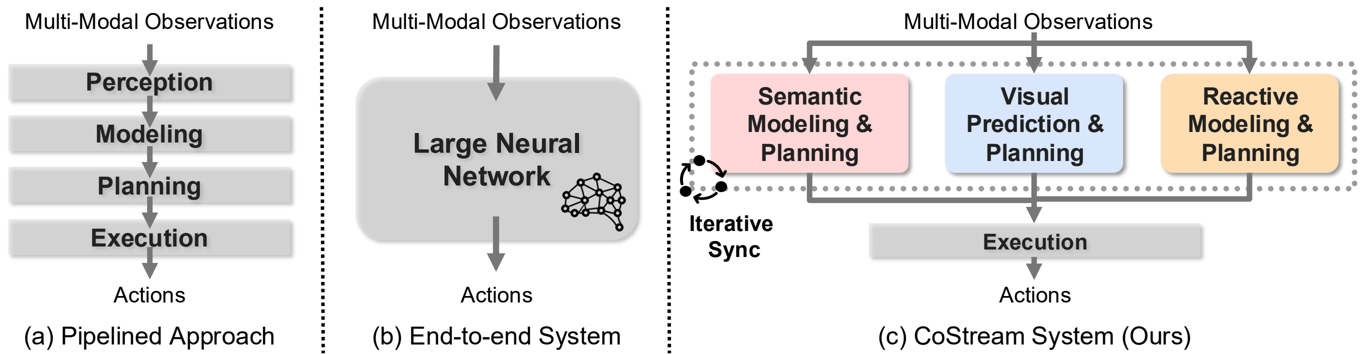


Fig. 1: **Comparison of Manipulation Approaches. (a) Classical Pipelines** use hand-engineered modules (perception, modeling, planning) that require extensive re-specification for new tasks. **(b) End-to-End Systems** train monolithic policies that map language and observations to actions but struggle with compositional generalization. **(c) `CoStream` (Ours)** uses multiple computational models operating on multi-sensory inputs to generate different components of task behaviors: a Semantic Modeling and Planning stream uses LLMs and VLMs to generate constraints for motion planning, a Visual Prediction and Planning stream uses video generation world models to generates the trajectories, and a Reactive Modeling and Planning stream incorporates tactile and force feedback to provide real-time compliant corrections during physical contact. The final control action is a unified transformation synthesized from all three streams.

*Abstract*—**Long-horizon contact-rich manipulation tasks demand that robots bridge the gap between slow, semantic planning and high-frequency, reactive control. Classical pipelines struggle with robust deployment due to brittle, hand-engineered interfaces, whereas monolithic end-to-end policies lack the flexibility to recombine learned skills for new scenarios. We propose a multi-stream architecture that combines foundation models (LLMs, VLMs, world models) with diverse sensing modalities. These streams function as autonomous, parallel processes that generalize through complementary, non-blocking mechanisms: the semantic stream applies abstract reasoning for task decomposition and constraint specification, the visual imagination stream utilizes world models to generate trajectories, and the reactive stream generates high-frequency compliant correction behaviors. These outputs are integrated into a final action through the multiplication of object-centric transformations. We validate our approach on long-horizon contact-rich assembly tasks such as drill insertion and motherboard assembly including RAM, CPU, and GPU components. Our results show that this method substantially outperforms prior work and demonstrates robust recovery against external human perturbations during execution.**

## I. INTRODUCTION

Robots performing long-horizon fine-grained manipulation must operate across different timescales and levels of abstraction [20, 43]. Consider a robot tasked with assembling computers by inserting CPUs, GPUs, and RAM sticks. Semantically, it must understand component-specific constraints. Geometrically, it must plan collision-free trajectories while maintaining precise relative object orientations. Reactively, it must regulate contact forces based on force and tactile feedback. These levels form a natural hierarchy spanning abstraction levels and time: slow deliberate planning transitions to fast reactive control. How can we build systems that handle this multi-level complexity? Two dominant paradigms, each with fundamental limitations, have emerged in robotics.

**Classical pipeline approaches** decompose manipulation into hand-engineered modules for perception, modeling, planning, and control [10, 41, 42]. Although these approaches are modular, they fail to generalize and must be manually specified for new tasks. Even for tasks that share similar alignment and insertion mechanics, like GPU installation and RAM insertion, these approaches require re-engineering.

**End-to-end learned approaches** avoid this brittleness by training monolithic policies that map directly from language and observations to actions. Recent work has explored vision-language-action models [6, 27, 38] and behavior cloning [13, 14, 18, 39]. Nevertheless, these approaches struggle with recombining learned knowledge for new scenarios, or compositional generalization. As they do not decompose problems into semantic, planning, and reactive components, such policies are unable, for instance, to transfer computer assembly knowledge across component types and part configurations.

We propose `CoStream`, a multi-stream framework that addresses these limitations. Our key insight is that *generalizable manipulation requires matching computational models to sensor modalities for different task behaviors*. By decomposing tasks into *parallel* behavior layers with clear interfaces, our architecture enables compositional generalization through layer reuse.

As illustrated in Fig. 1, **CoStream** consists of three non-blocking, parallel layers. The first layer uses LLMs and VLMs to decompose language instructions into semantic constraints that are optimized to yield high-level motion plans. A second visual imagination layer utilizes world models to produce object-centric trajectories by imagining candidate executions and tracking 3D keypoints. Simultaneously, the final reactive layer uses tactile and force sensors to provide high-frequency corrective behaviors during contact. Outputs from these layers are integrated into a final control signal by composing object-centric transformations. This composition mechanism allows the system to maintain closed-loop stability and asynchronous reactivity even when high-level semantic reasoning or visual imagination layers introduce large computational latency.

This design also achieves compositionality by reusing modules across different tasks. Each layer handles its respective abstraction level – semantic, trajectory generation, or reactive – while interfacing through well-defined inputs and outputs. In contrast, classical pipelines require manual extensive re-engineering for compositionality, while end-to-end policies need new demonstrations and model retraining.

We validate **CoStream** on long-horizon contact-rich assembly tasks including drill insertion and motherboard assembly of RAM, CPU, and GPU components. In these tasks, our approach substantially outperforms prior methods and demonstrates robustness through its ability to recover from external human perturbations during execution.

Our primary contributions are:

1) A multi-stream architecture combining foundation models (LLMs, VLMs, video generation) with complementary modalities (vision, language, tactile, force) to achieve compositional generalization across manipulation tasks.
2) A reactive modeling and planning layer that uses tactile and force feedback to enable transfer across contact-rich manipulation tasks without retraining.
3) A comprehensive evaluation of the system's capabilities in high-precision, long-horizon sequences including drill insertion and motherboard component assembly.

## II. Related Works

### A. Foundation Models for Robotics

Large Language Models (LLMs) and Vision-Language Models (VLMs) have enabled robots to interpret natural language and reason about long-horizon tasks [1, 6, 16, 22, 24]. Early approaches, such as Code as Policies [31], leveraged LLMs to synthesize executable Python programs that process perception outputs. To improve spatial precision, recent methods have shifted toward visual grounding. Works such as VoxPoser [28] compose value maps for planning, while ReKep [29], MOKA [17], and RoboPoint [50] prompt VLMs to predict specific keypoint constraints or affordances. Similarly, PIVOT [35] employs iterative visual prompting to refine action proposals directly in image space. Recognizing that visual reasoning often lacks physical understanding, recent work has attempted to incorporate physics into the planning loop. SIMPACT [34] and Prompting with the Future [36] use test-time simulation or digital twins to forecast and evaluate action outcomes. In general, these approaches rely on vision and predictive simulation. As a result, they lack the real-time tactile and force feedback required for contact-rich manipulation, where success depends on real-world interaction dynamics that neither vision nor simulation can reliably capture.

### B. Tactile and Force Sensing in Robotics

While vision is central to robotic perception, touch captures the physical interaction details required for contact-rich tasks like cable routing, packing, or needle threading [2, 21, 40, 51]. To address this gap, existing research generally falls into two categories. The first consists of analytic or constraint-based methods. These systems typically use tactile signals as mathematical constraints for optimization controllers [37] or as triggers for hand-designed motion primitives [47]. However, because these methods rely on explicit physical models or task-specific heuristics, they often require significant manual engineering for each new scenario. The second category employs data-driven learning methods. This includes reinforcement learning (RL) [23, 49] and imitation learning (IL) [5, 11, 25, 32, 48], as well as model-based methods that learn tactile-informed dynamics for model predictive control (MPC) [2, 44]. While these approaches can master complex skills by learning from demonstrations or interaction data, they are inherently data-hungry and tend to produce specialized policies that need retraining to transfer to new tasks. We propose an approach that leverages tactile feedback for object-centric behaviors via iterative optimization. Unlike rigid analytic models or specialized learned policies, our framework enables real-time adaptation and zero-shot generalization across novel objects and tasks without requiring task-specific training.

### C. Compositional Modeling for Robotics

Building generalizable robots requires compositionality, or the ability to efficiently recombine learned knowledge to solve new tasks. Generally, research has followed two distinct paradigms: traditional modular approaches or monolithic end-to-end policies. The former approaches, such as Task and Motion Planning (TAMP) [19] or layered control architectures [8], offer logical structure at the cost of brittleness. They typically require re-specification of interfaces and constraints for every new scenario. Conversely, monolithic end-to-end policies represent a non-compositional alternative [6, 7, 12, 52]. They avoid manual engineering, but typically learn entangled mappings that struggle with compositional generalization, and need extensive retraining to adapt to new tasks. To address these limitations, recent works have explored policy composition. Approaches like PoCo [45] aggregate policies from heterogeneous sources, while others explore factorized [33] or multimodal [11] composition for multi-task learning. However, these methods generally operate at a single level of abstraction by combining similar motion policies. They do not explicitly structure the system into distinct functional layers. Another line of work [3, 15] combines foundation models to hierarchically refine plans from abstract text to visual sequences, then it uses learned inverse models to infer robot actions. Our method draws inspiration from the layered philosophy of [8] but achieves hierarchical composition by integrating semantic, visual, and reactive layers. Unlike sequential models, our framework derives parallel, non-blocking objectives from each stream to drive robot behavior. Further, by recombining these modules for novel tasks, our framework avoids both the manual engineering of TAMP and the retraining requirements of end-to-end policies.

## III. METHODOLOGY

### A. Problem Formulation

We formulate long-horizon, contact-rich manipulation as a decision-making problem within a partially observable dynamical system. Let $\mathcal{S}$ denote the full state space and $\mathcal{O}$ the observation space. We assume the robot has access to the 6D poses of all task-relevant objects, denoted as $\mathcal{X}_t = \{\mathbf{T}_1, \ldots, \mathbf{T}_M\} \in SE(3)^M$. While the global geometric state $\mathcal{X}_t$ is assumed to be known, the local contact dynamics (e.g., friction coefficients, surface stiffness) and transient disturbances (e.g., in-hand slippage) remain unknown.

At each time step $t$, the robot receives a multimodal observation $o_t = (I_t, \mathcal{X}_t, F_t) \in \mathcal{O}$, where $I_t$ represents visual data (RGB-D), $\mathcal{X}_t$ denotes the object poses, and $F_t$ denotes tactile/force feedback. The robot is tasked with a high-level natural language instruction $\mathcal{L}$. Our objective is to learn a composite policy $\pi(a_t \mid o_{0:t}, \mathcal{L})$ that generates a sequence of commands $a_t = (\mathbf{T}_{ee,t}, g_t)$, where $\mathbf{T}_{ee,t} \in SE(3)$ is the target pose and $g_t \in [0, 1]$ is the gripper width, to maximize the probability of task success. We decompose this policy into three concurrent processes, each operating at a distinct level of abstraction to drive complementary task behaviors:

1) A Semantic Modeling and Planning layer $\pi_{\text{plan}}(\{\mathcal{C}^k, \mu^k\}_{k=1}^N \mid \mathcal{L}, I_t, \mathcal{X}_t)$ that decomposes the task into $N$ stages, mapping instructions to geometric constraints $\mathcal{C}^k$ and determining the reactive control strategy $\mu^k$ (stiffness and compensation configurations).
2) A Visual Trajectory Generation layer $\pi_{\text{traj}}(\tau_{\text{ref}}^k \mid \mathcal{C}^k, I_t)$ that leverages a video world model to "imagine" the execution, synthesizing a physically plausible object-centric reference trajectory $\tau_{\text{ref}}$.
3) A Reactive Modeling and Planning layer $\pi_{\text{react}}(a_t \mid \tau_{\text{ref}}, F_t, \mu^k)$ that utilizes the active compliance strategy $\mu^k$ to modulate a dual-objective reflex loop, optimizing end-effector poses to reject slippage (detected via $I_{\text{tac}}$) while yielding to jamming forces (detected via $F_t$).

Fig. 2 summarizes **CoStream**'s architecture. In the following sections, we introduce each layer in more detail.

### B. Semantic Modeling and Planning

The semantic planning layer converts high-level task instructions into geometric objectives and reactive control strategies. It does not generate trajectories or low-level control signals directly; instead, it defines what constraints must be met and how the robot should interact physically.

*1) Task decomposition & capabilities:* We use a language model to parse the instruction $\mathcal{L}$ into an ordered sequence of semantic stages $\{S^1, \ldots, S^N\}$. For each stage, a VLM identifies task-relevant keypoints and outputs a set of geometric constraints $\mathcal{C}^k$ (e.g., alignment costs defined in Eq. 1).

*a) Geometric constraints:* A Vision-Language Model (VLM) processes the current observation $I_t$ to identify task-relevant keypoints. We represent the spatial goals as cost functions over the task space. For example, an alignment objective is defined as:

$$\mathcal{C}_{\text{align}}(\mathbf{T}_{ee}) = \|\mathbf{T}_{ee}\mathbf{p}_{\text{tool}} - \mathbf{T}_{\text{obj}}\mathbf{p}_{\text{target}}\|_2^2,$$

where $\mathbf{T}_{ee} \in SE(3)$ is the end-effector pose, and $\mathbf{p}_{\text{tool}}, \mathbf{p}_{\text{target}} \in \mathbb{R}^3$ are local keypoint offsets defined in the tool and target object frames, respectively.

*b) Compliance strategy:* Simultaneously, the VLM generates a stage-specific contact strategy $\mu^k = \{K^k, S_{\text{tac}}^k\}$ that governs the robot's physical interaction behavior:

- **Stiffness Matrix ($K^k$):** A diagonal gain matrix defining the Cartesian impedance. This modulates the system's passive compliance, enforcing high rigidity during free-space transport to reject disturbances while lowering stiffness during contact phases to safely accommodate geometric uncertainty.
- **Selection Matrix ($S_{\textbf{tac}}^k$):** A binary diagonal matrix that partitions the task space into *active correction axes* and *passive drift axes*. For axes where $S_{ii} = 1$, the controller actively compensates for tactile error to enforce alignment; for axes where $S_{ii} = 0$, the controller ignores tactile drift, allowing environmental constraints (such as a hole's depth) to dictate the object's pose.

*2) The semantic solver:* Once the constraints $\mathcal{C}^k$ are defined, we solve for a nominal end-effector pose $\mathbf{T}_{\text{nom}}^* \in SE(3)$ by minimizing a weighted sum of the objectives:

$$\mathbf{T}_{\text{nom}}^* = \arg\min_{\mathbf{T}\in SE(3)} \sum_{j=1}^{M} w_j \mathcal{C}_j(\mathbf{T}, \mathcal{X}_t),$$

where $\mathcal{C}_j$ denotes stage-specific objectives (e.g., alignment or approach constraints), $\mathcal{X}_t$ summarizes the current scene estimate, and $w_j$ are scalar weights.

To solve the objective function, we first apply Basin Hopping to obtain an initial solution and then refine it using Sequential Least Squares Programming (SLSQP). The solver is warm-started across timesteps to maintain continuity.

### C. Visual Trajectory Generation

To bridge the gap between static semantic plans and dynamic execution, we employ a visual imagination pipeline. Instead of relying on a single deterministic plan, we generate multiple candidate futures, select the optimal one using a VLM, and extract an actionable 3D trajectory.

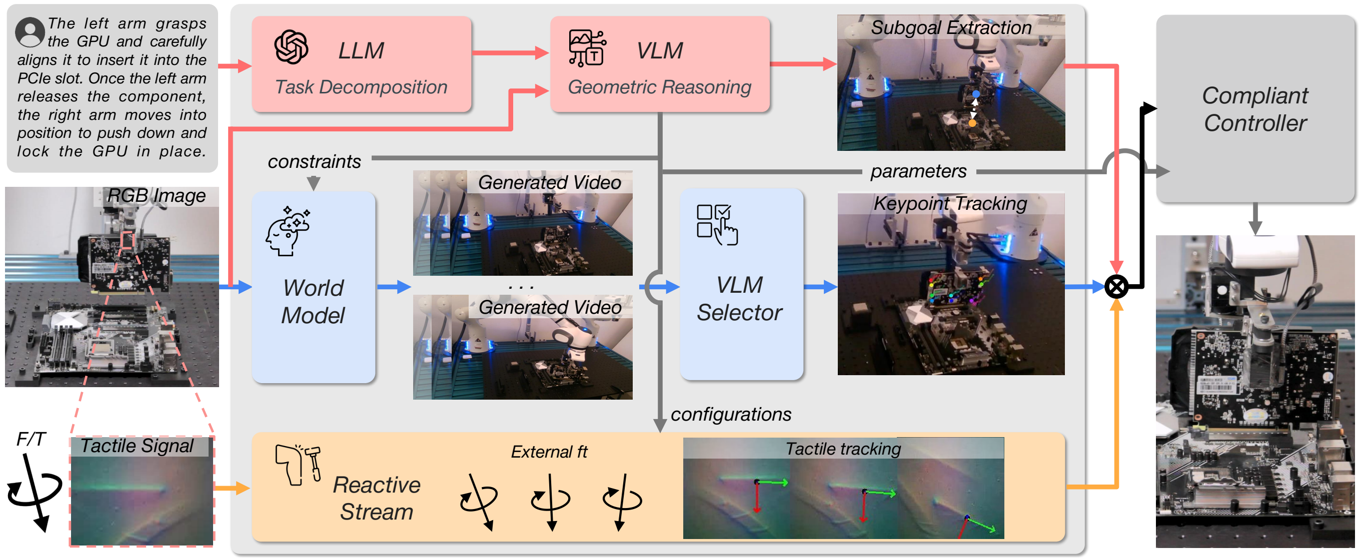


Fig. 2: **CoStream Architecture.** **CoStream** consists of three concurrent computational streams. The Semantic Modeling and Planning layer infers constraints derived from language goals and visual observations. The Visual Trajectory Generator leverages world models to "dream" multiple candidate execution videos, uses a VLM to select the optimal sequence, and extracts 3D trajectories via keypoint tracking. The Reactive Modeling and Planning layer generates high-frequency tactile/force-torque reflex behaviors to maintain local contact stability during execution.

*1) Video imagination & VLM selection:* We leverage the Large Video Planner (LVP) [9] as a world model to "imagine" the future. We condition the model on the current observation $I_t$ and the text instruction $\mathcal{L}_{\text{step}}$. To ensure diversity, we sample $K$ distinct video rollouts $\{V_1, \dots, V_K\}$, each predicting a potential visual outcome of the robot's action over a short horizon.

A VLM-based critic then evaluates these candidates to filter out physically implausible or unsafe trajectories (e.g., collisions or misalignment) before they are executed. Given the instruction $\mathcal{L}_{\text{step}}$ and the set of generated videos, the VLM selects the optimal video $V^*$ that best satisfies both the semantic and safety constraints.

*2) Trajectory extraction via keypoint tracking:* Given the optimal video $V^*$, we then convert the 2D visual plan into a 3D control trajectory. We employ a 3D keypoint tracker to trace the motion of task-relevant objects within the imagined video [46]. We lift each 2D track into 3D space using the initial camera intrinsics and the depth information, propagating the depth through the generated sequence.

This process yields an object-centric reference trajectory $\tau_{\text{ref}} = \{\mathbf{p}_{3\text{D}}(t)\}_{t=1}^{T}$. This trajectory guides the robot's motion, ensuring it follows the "imagined" path validated by the VLM.

### D. Reactive Modeling and Planning

The Reactive Layer operates as a high-frequency (30 Hz) reflex loop. To ensure both precision and safety, we implement a Dual-Objective Reactive Controller that combines two complementary reflex loops: one for geometric consistency (from tactile feedback) and one for physical safety (from external force-torque).

*1) Tactile pose tracking:* We utilize the GelSight Mini sensor to detect high-frequency relative motion between the end-effector and the object. We employ the inverse compositional NormalFlow algorithm [26] to solve for the incremental warp.

To actively reject unmodeled disturbances, we adapt the NormalFlow [26] algorithm as a high-frequency state estimator. We utilize the inverse compositional formulation to ensure real-time performance [4]. Unlike standard alignment approaches that re-evaluate gradients at every step, the inverse compositional formulation allows us to pre-calculate the Hessian matrix for the reference template. We solve for an incremental warp parameter $\Delta\xi$ that aligns the current observation to the reference:

$$\Delta\xi^* = \arg\min_{\Delta\xi} \sum_{\mathbf{u}\in\Omega} \left\| R(\Delta\xi)^{-1} I_{\text{obs}}(W(\mathbf{u};\xi)) - I_{\text{ref}}(\mathbf{u}) \right\|^2,$$

where $\Omega$ denotes the set of pixel coordinates within the contact region, $I_{\text{obs}}, I_{\text{ref}}$ are the observed and reference normal maps, and $W(\mathbf{u};\xi)$ is the warp function mapping pixel coordinates $\mathbf{u}$ according to the twist $\xi$.

Since surface normal maps are scale-ambiguous regarding depth, we decouple the $z$-axis translation and solve for it analytically. We compute the vertical correction $\Delta z$ by averaging the depth discrepancy over the contact patch:

$$\Delta z = \frac{1}{|\Omega|} \sum_{\mathbf{u}\in\Omega} \left[ D_{\text{obs}}(W(\mathbf{u};\xi^*)) - \left(\mathbf{R}_{\xi^*} \cdot \mathbf{q}(\mathbf{u}) + \mathbf{t}_{\xi^*}\right)_z \right]$$

where $D_{\text{obs}}$ is the depth map derived via Poisson integration, $\mathbf{q}(\mathbf{u}) \in \mathbb{R}^3$ is the unwarped 3D surface point at pixel $\mathbf{u}$, and $(\mathbf{R}_{\xi^*}, \mathbf{t}_{\xi^*})$ are the rotation and translation components of the estimated twist $\xi^*$.

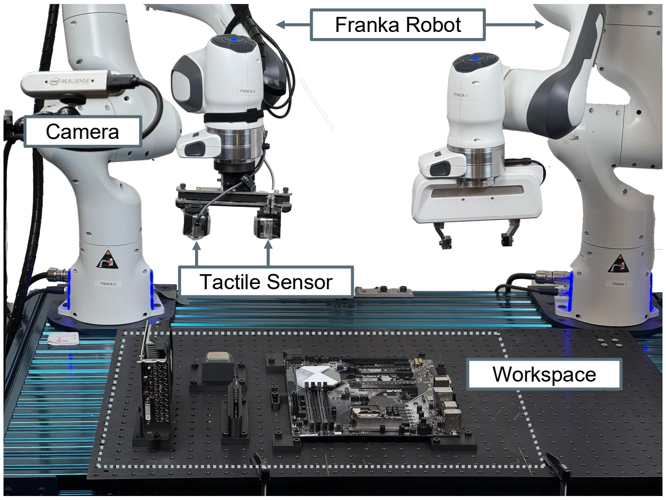


Fig. 3: **Experimental Setup.** The system utilizes two Franka Emika Panda robots. One manipulator is equipped with a GelSight tactile sensor. The dashed outline indicates the workspace, while a side-mounted camera provides visual perception of the environment.

Finally, we map these optimized parameters into a cumulative slip transform $\Delta\mathbf{T}_{\text{slip}}$ via the exponential map. This transform acts as the error term for our Reactive Execution Layer:

$$\Delta\mathbf{T}_{\text{slip}} \leftarrow \Delta\mathbf{T}_{\text{slip}} \cdot \exp(\Delta\xi^*)^{-1} \tag{1}$$

*2) Iterative alignment & composite control:* To maintain stable manipulation, the robot must actively compensate for the detected slip. We treat the slip estimate $\Delta\mathbf{T}_{\text{slip}}$ from Eq. (1) as a drift error that must be zeroed out.

The final control command is generated by superimposing this high-frequency correction onto the low-frequency nominal plan $\mathbf{T}^*_{\text{nom}}$:

$$\mathbf{T}_{\text{cmd}} = \mathbf{T}^*_{\text{nom}} \oplus (\lambda_{\text{gain}} \cdot \Delta\mathbf{T}_{\text{slip}})$$

where $\oplus$ denotes rigid body composition in the end-effector frame, and $\lambda_{\text{gain}} \in [0, 1]$ is a scalar gain that modulates the aggressiveness of the correction.

This creates a closed-loop “chasing” behavior: as the object slips or rotates in hand, the end-effector continuously realigns itself to the local surface frame, ensuring the geometric constraints of the global plan remain valid relative to the object.

## IV. EXPERIMENTS

To validate our framework, we conducted real-world experiments on long-horizon, high-precision robotic manipulation tasks. Our evaluation aims to address three core research questions: (**RQ1**) whether the asynchronous composition of reactive modeling and planning effectively rejects real-time perturbations to maintain sub-millimeter precision; (**RQ2**) whether the multi-layered architecture enables zero-shot transitions between tasks with diverse physical constraints without retraining; and (**RQ3**) if the integration of a world model for trajectory “imagination” provides physically consistent motion priors that improve execution stability.

### A. Experimental Setup

*Hardware:* We utilize a robotic setup with two Franka Emika Panda arms. One arm is equipped with a GelSight Mini tactile sensor for precision manipulation, while the other uses a standard parallel-jaw gripper. External perception is provided by calibrated RealSense D415 cameras. Fig. 3 depicts the robots and sensor suite.

*Tasks:* We evaluate our framework on four real-world manipulation tasks inspired by desktop computer assembly. We conduct two types of evaluations: (1) *Individual subtasks* ($N = 15$ trials each) to statistically analyze precision and robustness, and (2) *Full assembly* where the subtasks are executed sequentially to validate system integration.

1) *Robotic Drill Insertion:* One arm holds the drill steady while the other inserts the bit into a hole with a tiny 0.5 mm gap. This requires the robot to keep the bit perfectly centered or cause the drill to jam immediately.
2) *RAM Insertion:* The robot inserts a RAM module into a slot. This task requires precise alignment, followed by a high force to click the module into the locked position.
3) *CPU Placement:* The robot places a CPU chip into its socket. Unlike the RAM task, this requires a very gentle touch. The chip must be perfectly flat and aligned to avoid bending the delicate pins.
4) *GPU Insertion:* The robot slides a graphics card into a PCIe slot. Because the card is long, even a tiny amount of tilt can cause it to get stuck during insertion.

### B. Baselines

We compare **CoStream** against two state-of-the-art paradigms that represent the current landscape of foundation-model-based robotics:

- **VoxPoser [28]:** VoxPoser adopts a *modular pipeline approach* using foundation models. It uses LLMs and VLMs to generate 3D affordance and value maps that guide a motion planner. VoxPoser represents a class of single-stream pipeline architectures. This comparative evaluation tests the benefit of our multi-stream, multisensory framework over systems that rely on purely geometric planning without high-frequency reactive feedback.
- $\pi_{0.5}$ **[30]:** This baseline represents a *monolithic end-to-end approach*. It is a large-scale Vision-Language-Action (VLA) model that maps observations directly to continuous control commands. This comparison validates whether our decomposed architecture provides superior precision and compositional flexibility in contact-rich scenarios compared to large, unified neural network policies.

### C. Results and Analysis

*1) Quantitative success rates:* Table I summarizes performance across 15 trials per task. **CoStream** consistently outperforms baselines, particularly in tasks requiring force awareness. As shown in the results, our **CoStream** framework achieves high success rates across all tasks while the baselines do not complete any trials successfully.

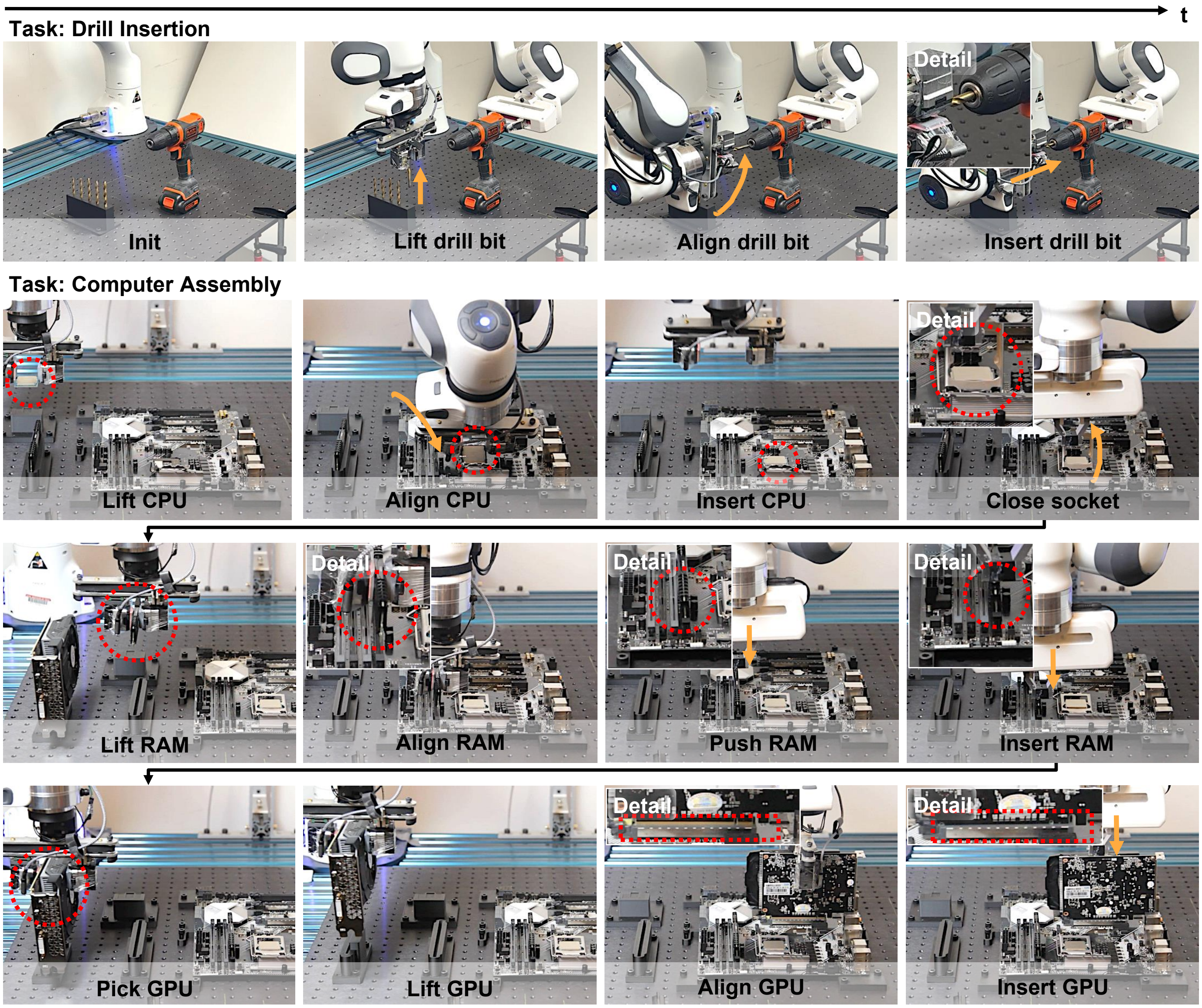


Fig. 4: **Qualitative Rollouts of Contact-Rich Manipulation. (Top) Drill Insertion:** The system performs high-precision insertion with a 0.5 mm clearance. Our reactive layer enables sub-millimeter compliance and corrections in real-time, maintaining contact to prevent jamming. **(Bottom) Computer Assembly Sequence:** The system is able to sequentially execute CPU placement, RAM insertion, and GPU installation. By updating object-centric constraints, the same architecture transitions from delicate, force-sensitive CPU handling to the high-force requirements of RAM locking without task-specific retraining.

*2) Failure analysis of baselines:* The failure of VoxPoser and $\pi_{0.5}$ highlights the inherent difficulties of contact-rich manipulation. Both baselines operate primarily on visual feedback and lack the necessary mechanisms to handle the precision requirements of these tasks.

For VoxPoser, the system proved brittle as a break in any pipeline subcomponent led to total system failure. The model failed to generate accurate affordance maps for contact-rich insertion, resulting in trajectories that did not align with the physical constraints of the assembly components. This lack of physical grounding in the high-level reasoning chain prevented the robot from reaching the initial alignment phase.

For $\pi_{0.5}$, the robot frequently exhibited confusion during the execution phase. Although the policy initiated movement, it often resulted in jerky behavior in free space. This suggests that the monolithic VLA approach struggles to maintain the smooth, stable control necessary for sub-millimeter alignment in unstructured environments.

**CoStream** succeeds by treating the global plan as a nominal target. The visual trajectory generation layer utilizes a world model to produce smooth reference paths, which the reactive modeling and planning layer then continuously refines. This architecture enables the system to achieve long-horizon precise manipulation tasks with a high success rate.

*3) Qualitative analysis of rollouts (RQ1 & RQ2):* We visualize the execution capabilities of **CoStream** through

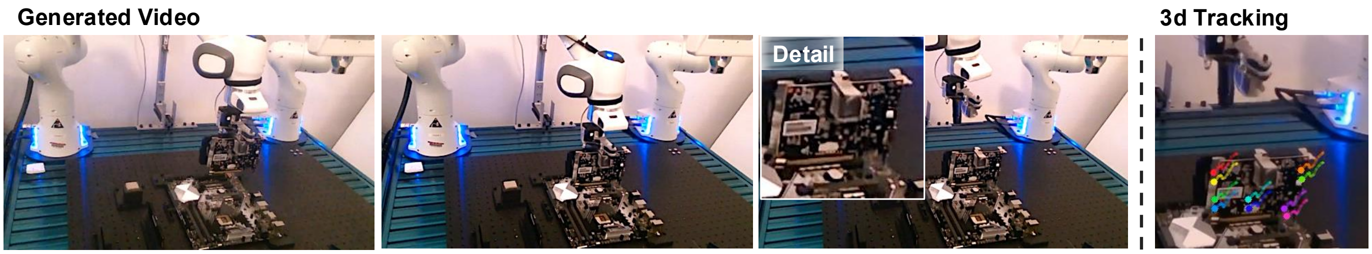


Fig. 5: **Visual Trajectory Generation and Keypoint Tracking.** The left part illustrates the video frames generated by the world model (e.g., GPU insertion), while the right figure depicts the extracted 3D keypoint tracks. These imagined futures provide a high-fidelity motion prior for deriving smooth, object-centric trajectories.

TABLE I: **Quantitative Success Rates.** Performance comparison against modular (*VoxPoser*) and monolithic ($\pi_{0.5}$) baselines (15 trials each). Our multi-stream architecture achieves a 96.7% mean success rate and manages sub-millimeter clearances (0.5 mm) and complex contact dynamics that the baselines cannot resolve.

| Method | Drill Insertion | RAM Insertion | CPU Placement | GPU Insertion | Mean Success |
|---|---|---|---|---|---|
| VoxPoser | 0/15 | 0/15 | 0/15 | 0/15 | 0.0% |
| $\pi_{0.5}$ | 0/15 | 0/15 | 0/15 | 0/15 | 0.0% |
| **Ours** | **15/15** | **14/15** | **14/15** | **15/15** | **96.7%** |

policy rollouts and world model predictions. Fig. 4 illustrates a successful trajectory for the drill insertion task; the robot demonstrates compliant behavior, adapting its pose to the mechanical constraints of the hole. This reactive adjustment ensures the bit remains centered despite the narrow 0.5 mm clearance, preventing the jamming observed in baselines. As further shown in Fig. 4, we evaluate on a continuous assembly sequence (CPU, RAM, GPU) in a single rollout. Notably, the framework performs all these tasks without any retraining. This zero-shot transfer is made possible by the parallel compositionality of the architecture, allowing us to reuse semantic modeling, visual imagination, and reactive planning layers by simply updating object-centric constraints. This modularity allows the system to transition between tasks with vastly different physical properties – moving from delicate CPU handling to high-force RAM locking – without modifying the underlying control loops.

*4) Visual trajectory generation (RQ3):* Fig. 5 displays the output of the visual trajectory generation layer, where the left part represents the generated video and the right part shows the tracked keypoints. The world model effectively "imagines" robot trajectories. These generated videos serve as a reference for the motion prior; by extracting 3D keypoint tracks from these imagined futures, we derive a smooth object-centric trajectory and natural nominal plan.

*5) Robustness to human perturbation (RQ1):* Our framework exhibits significant resilience to external physical disturbances. We validate this by introducing manual perturbations during insertion, rotating the object within the gripper to simulate slippage or interference. As illustrated in Fig. 6, the reactive modeling and planning layer immediately identifies the deviation between the actual state and the referenced trajectory. In response, the controller dynamically reorients the end-effector to realign the object pose. Fig. 6 (left) demonstrates this recovery across RAM, CPU, and GPU insertion

TABLE II: **Ablation Study on Drill Insertion.** Removing the reactive layer yields a significantly lower success rate, meaning that force signal is essential for sub-millimeter precision.

| Method | Success Rate |
|---|---|
| Ours (Full Framework) | **15/15** |
| Ours w/o Reactive Layer | 3/15 |

tasks. The detailed breakdown (Fig. 6, right) highlights the system's ability to handle a diverse morphological spectrum, spanning the thin contact surfaces of a RAM module, the miniature footprint of a CPU, and the substantial mass of a GPU, maintaining high-precision alignment despite stochastic human interference.

*6) Ablation: Role of reactive feedback (RQ1):* To evaluate the specific contribution of the reactive layer, we performed an ablation study on the *Drill Insertion* task, denoted as *Ours w/o Reactive*. In the ablated setting, the robot follows the nominal trajectory but does not use high-frequency force and tactile feedback. The success rate for Drill Insertion dropped from 100% (15/15) to 20% (3/15) when the reactive layer was removed (Table II). Analysis of the failure modes revealed that reactive force sensing is critical for the system to handle the tight 0.5 mm clearance. The three successful trials occurred only when the initial global alignment was near-perfect; in all other instances, slight angular misalignments led to "jamming" or high-force spikes that the planner could not resolve.

## V. Conclusion and Future Works

In this work, we introduced **`CoStream`**, a multi-stream framework addressing the trade-offs between high-level semantic reasoning and fine-grained physical reactivity. By decomposing manipulation into parallel layers including VLM-based planning, world-model trajectory generation, and tactile-force control, we successfully bridge the gap between slow deliberate reasoning and fast precise execution. Our experiments

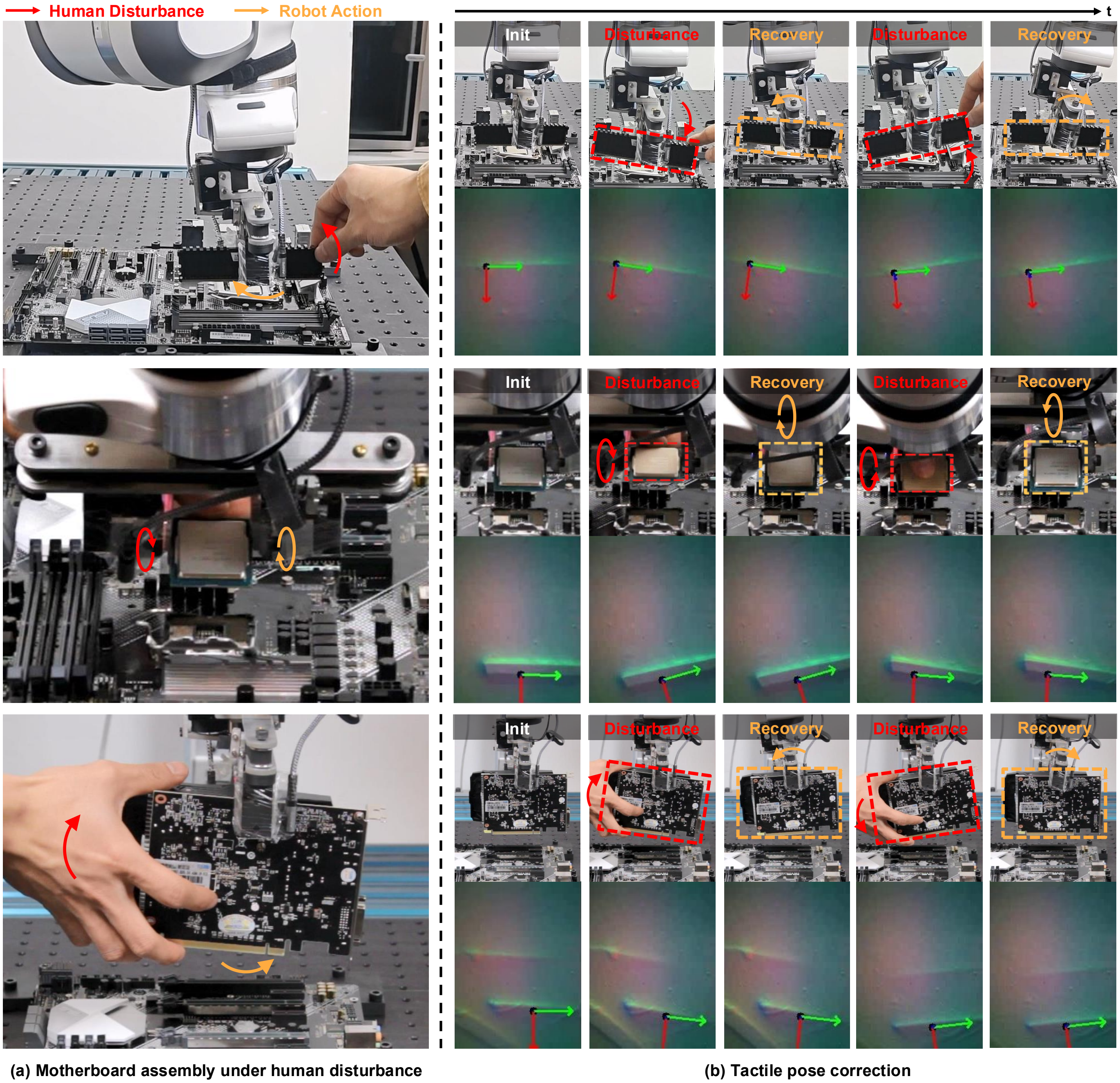


Fig. 6: **Robustness to Human Perturbation via Tactile Feedback.** (Left) The snapshots of the robot recovering from manual object displacement during the insertion of a RAM module (top), CPU (middle), and GPU (bottom). (Right) A detailed view of the reactive alignment for each component. The first row for each task illustrates the human-induced rotation of the object within the gripper fingers, while the second row visualizes the corresponding raw tactile sensor readings and the derived real-time pose estimations used by the reactive modeling and planning layer to maintain task consistency.

on complex, long-horizon assembly tasks demonstrate that this multi-stream approach significantly outperforms both classical pipeline approaches and monolithic large neural network baselines. By effectively grounding abstract VLM specifications into object-centric trajectories and sub-millimeter tactile corrections, our framework provides a scalable path for robots to plan, dream, and touch in complex real-world environments.

There are several promising avenues for future work. One primary direction involves reducing the inference latency of the video generation component to enable real-time, closed-loop trajectory updates that adapt to dynamic scene changes mid-execution. In addition, it would be interesting to explore the expansion of this architecture to multi-agent, multi-sensory scenarios, where coordinated streams act as specialized agents to execute increasingly complex, collaborative tasks.

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